particles, and that's good, since quantum mechanics describes the microworld with unprecedented accuracy. But for large objects such as experimenters and their equipment, which have billions and billions of particles, the odds are high that in a tiny fraction of any given second the posited spontaneous collapse will kick in for at least one constituent particle, causing its wavefunction to collapse. And, as argued by Ghirardi, Rimini, Weber, and others, the entangled nature of all the individual wavefunctions in a large object ensures that this collapse initiates a kind of quantum domino effect in which the wavefunctions of all the constituent particles collapse as well. As this happens in a brief fraction of a second, the proposed modification ensures that large objects are essentially always in one definite configuration: pointers on measuring equipment always point to one definite value; the moon is always at one definite location in the sky; brains inside experimenters always have one definite experience; cats are always either dead or alive.

Each of these approaches, as well as a number of others I won't discuss, has its supporters and detractors. The "wavefunction as knowledge" approach finesses the issue of wavefunction collapse by denying any reality for wavefunctions, turning them instead into mere descriptors of what we know. But why, a detractor asks, should fundamental physics be so closely tied to human awareness? If we were not here to observe the world, would wavefunctions never collapse, or, perhaps, would the very concept of a wavefunction not exist? Was the universe a vastly different place before human consciousness evolved on planet earth? What if, instead of human experimenters, mice or ants or amoebas or computers are the only observers? Is the change in their "knowledge" adequate to be associated with the collapse of a wavefunction?[11]

By contrast, the Many Worlds interpretation avoids the whole matter of wavefunction collapse, since in this approach wavefunctions don't collapse. But the price to pay is an enormous proliferation of universes, something that many a detractor has found intolerably exorbitant.[12] Bohm's approach also avoids wavefunction collapse; but, its detractors claim, in granting independent reality to both particles and waves, the theory lacks economy. Moreover, the detractors correctly argue, in Bohm's formulation the wavefunction can exert faster-than-light influences on the particles it pushes. Supporters note that the former